

Clock System in STM32F103C6

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Outline

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- 2 Clock Sources
- 3 Clock Distribution
- 4 Timer and Peripheral Clocks
- 5 Example Configuration
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Clock Configuration

pwm.ioc - Clock Configuration

Pinout & Configuration

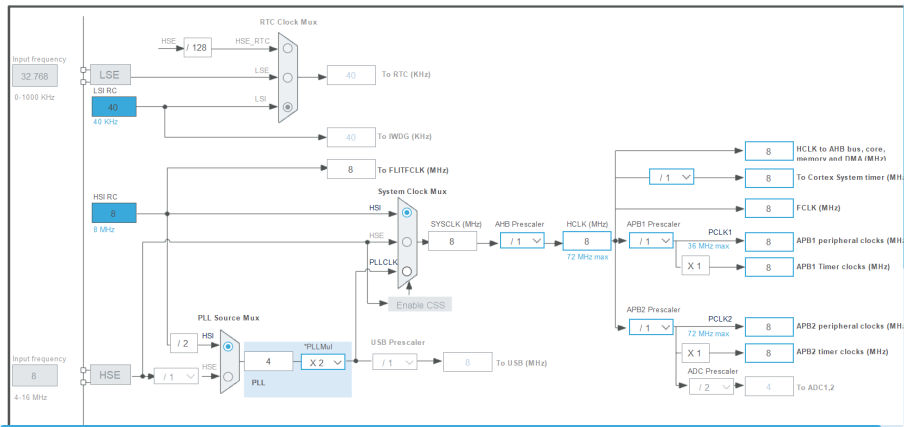
Clock Configuration

Project Manager

Tools



Resolve Clock Issues



Why Clock System?

- Provides synchronization for CPU and peripherals
- Determines performance and power consumption
- Multiple clock sources allow flexibility
- Must be configured properly for reliable operation

Main Clock Sources

- **HSI (High Speed Internal RC):** 8 MHz, internal, medium accuracy
- **HSE (High Speed External):** 4–16 MHz, external crystal or clock, high accuracy
- **PLL (Phase Locked Loop):** multiplies HSI/2 or HSE, up to 72 MHz
- **LSI (Low Speed Internal RC):** 40 kHz, for watchdog
- **LSE (Low Speed External):** 32.768 kHz, for RTC

System Clock Mux

- SYSCLK can be selected from: HSI, HSE, PLLCLK
- SYSCLK is distributed through prescalers to buses and peripherals

AHB and APB Prescalers

- SYSCLK → HCLK (AHB Prescaler)
- HCLK → APB1 (PCLK1) and APB2 (PCLK2)
- **APB1**: Max 36 MHz (low-speed peripherals)
- **APB2**: Max 72 MHz (high-speed peripherals)

- Timer1 belongs to APB2: Clock = PCLK2 or $2 \times \text{PCLK2}$ if prescaler $\neq 1$
- Timer2,3,4 belong to APB1: Clock = PCLK1 or $2 \times \text{PCLK1}$ if prescaler $\neq 1$
- ADC uses PCLK2 with additional divider (max 14 MHz)
- USB requires precise 48 MHz

Typical 72 MHz Setup

- HSE = 8 MHz
- PLL multiplier = 9 $\rightarrow$ SYSCLK = 72 MHz
- HCLK = 72 MHz
- APB2 = 72 MHz (Timer1 = 72 MHz)
- APB1 = 36 MHz (Timer2,3,4 = 72 MHz due to x2 rule)
- USB prescaler = 1.5 $\rightarrow$ 48 MHz

- 1 Configure SYSCLK from HSI = 8 MHz and measure SysTick delay.
- 2 Configure HSE = 8 MHz with PLL x9 to achieve 72 MHz.
- 3 Verify Timer PWM output frequency matches theoretical calculation.

Summary

- STM32F103C6 has flexible clock sources: HSI, HSE, PLL, LSI, LSE.
- SYSCLK distributed via AHB and APB prescalers.
- Timers may run at double PCLK when prescaler $\neq 1$.
- Correct configuration ensures performance and peripheral accuracy.